Introduction to Cybersecurity

Computer crime and digital evidence

Jasmin Azemović
CISO | Associate Professor | Microsoft MVP

Sections

- Basic terms
- Digital forensics requirements
- Collecting methods

Basic terms

Basic terms

- Computer crime
- Digital forensics
- Investigation proces
- Phases of digital forencis



Computer crime

- Computer crime is activity where computer is: resource, service, tool, target or place of criminal act.
- Goals?
  - Electronic fraud
  - Identity hijacking
  - Interferences of within the data
  - Unauthorized access;
  - Network interceptions;
  - …



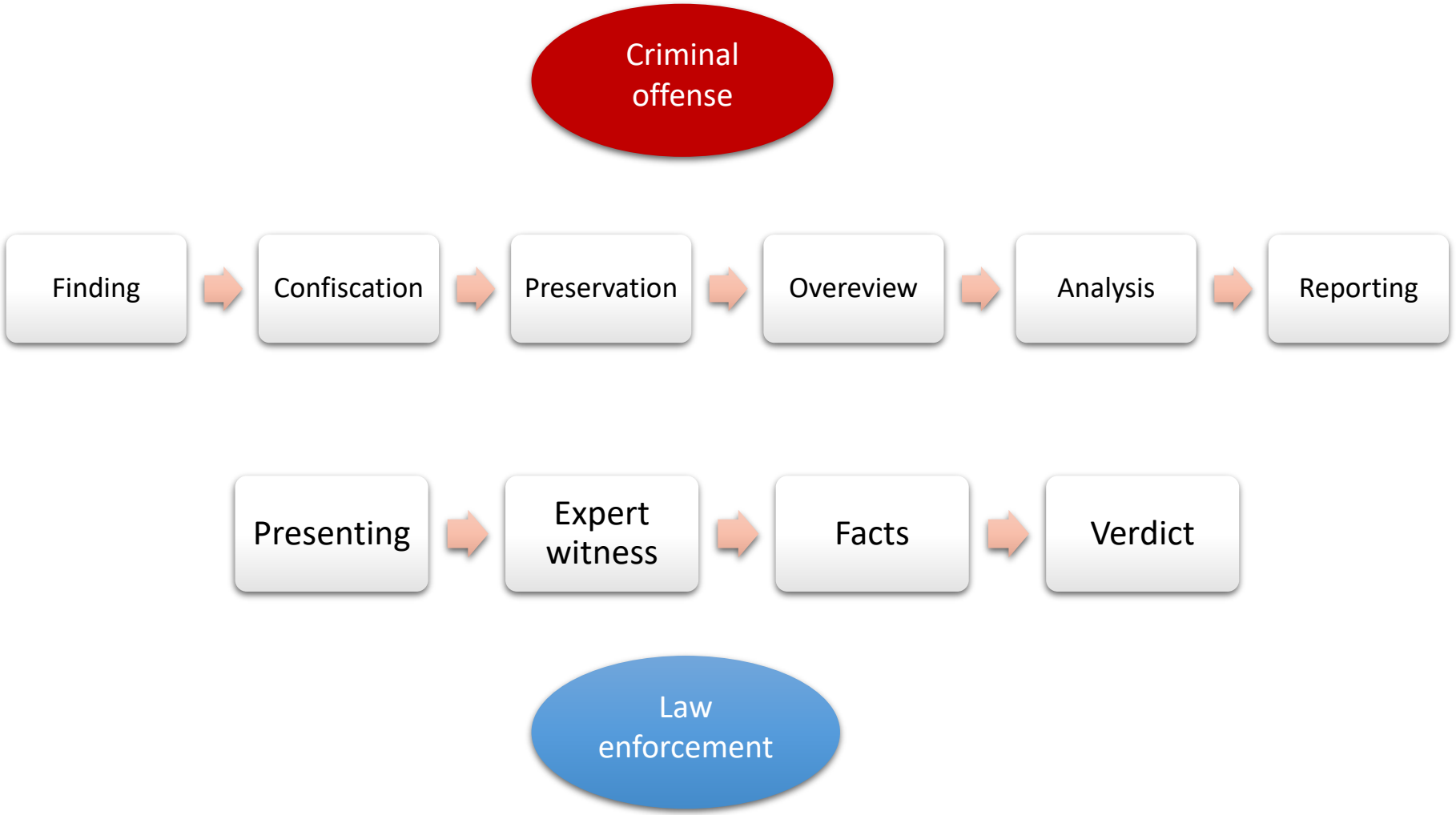
Digital forensics

- ...is a part of the investigation, which involves the scientific analysis: the media / storage devices and / or devices for data processing;
- Servers, PC, laptop, tablet, mobile devices, storage media...

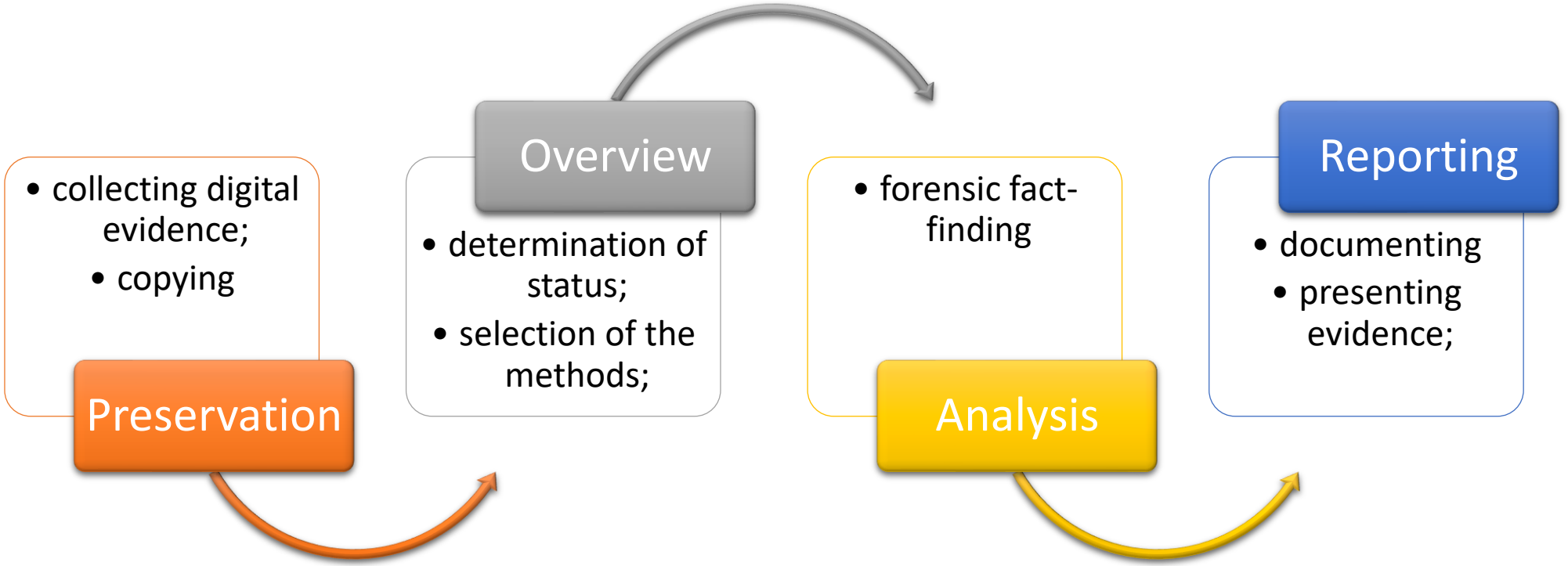
- ... with the aim of finding traces of the activities that led to the crime.



Investigation proces



Phases of digital forencis



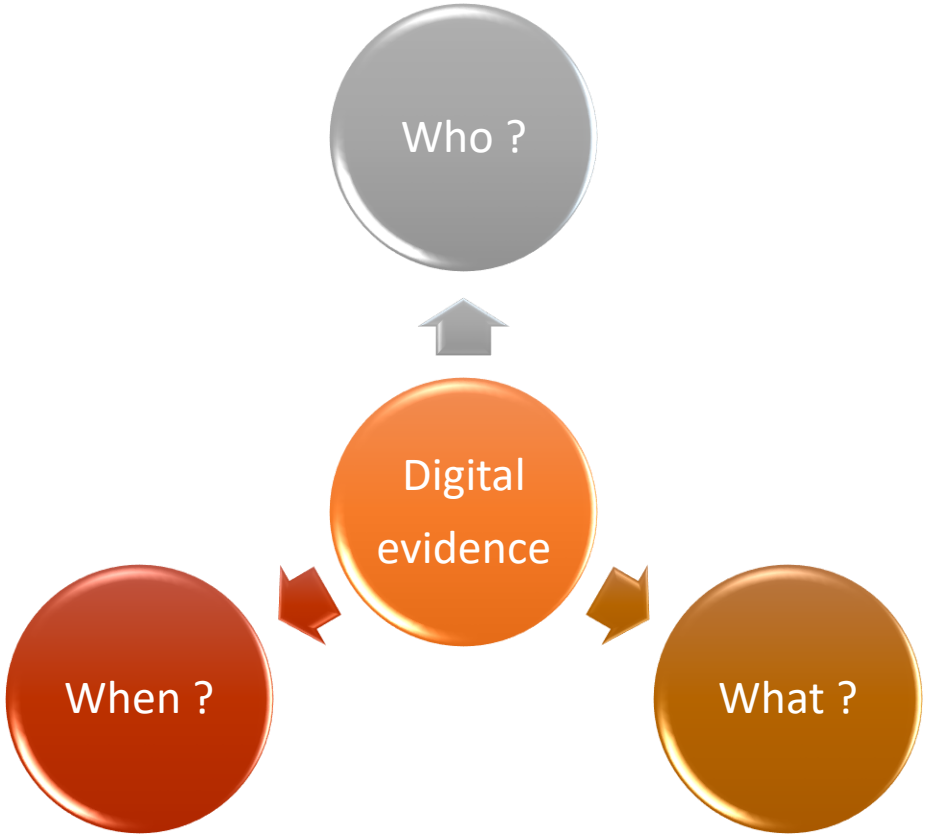
Digital forensics requirements

Digital forensics requirements

- Access control;
- Goals of access control;
- Simple access control;
- Advanced access control;

Access control

- The only way for a quality collection of digital evidence
- Independence from the client
- Goals



Access control

- The only way for a quality collection of digital evidence is the existence of an access control mechanism;
- Independence from the client (Dekstop GUI, Web environment, direct access ..);
  - Simple access control;
  - Advanced access control;

Simple access control

Results | Messages

	StudentID	Prezime	Ime	Datum	Url	RemoteIp	RemoteAgent	Referer
8	3486	ći	siv	2011-03-07 11:30:38.517	/nastava/index_predmet.aspx	10.0.0.1	Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; ...	/nastava/index_predmet.aspx
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Advanced access control

- Everything that has a simple access control, plus:
- Data on the DML event (INSERT, UPDATE and DELETE);
- Data on DDL events (CREATE, ALTER, DROP)
- Protection of audit records from tampering
  - With the aim of protecting the data collected from falsification, deletion or modification

Collecting methods

Collecting methods

- DDL triggers
- DML triggers
- AntiTampering
- Tamper detection model



DDL triggers

- Server and database levele events
- Accompanied by changes in the scheme of objects
  - CREATE, ALTER, DROP;
- *Eventdata* function collects data for the DDL event;
  - The output is a collection of data in XML format

DML triggers

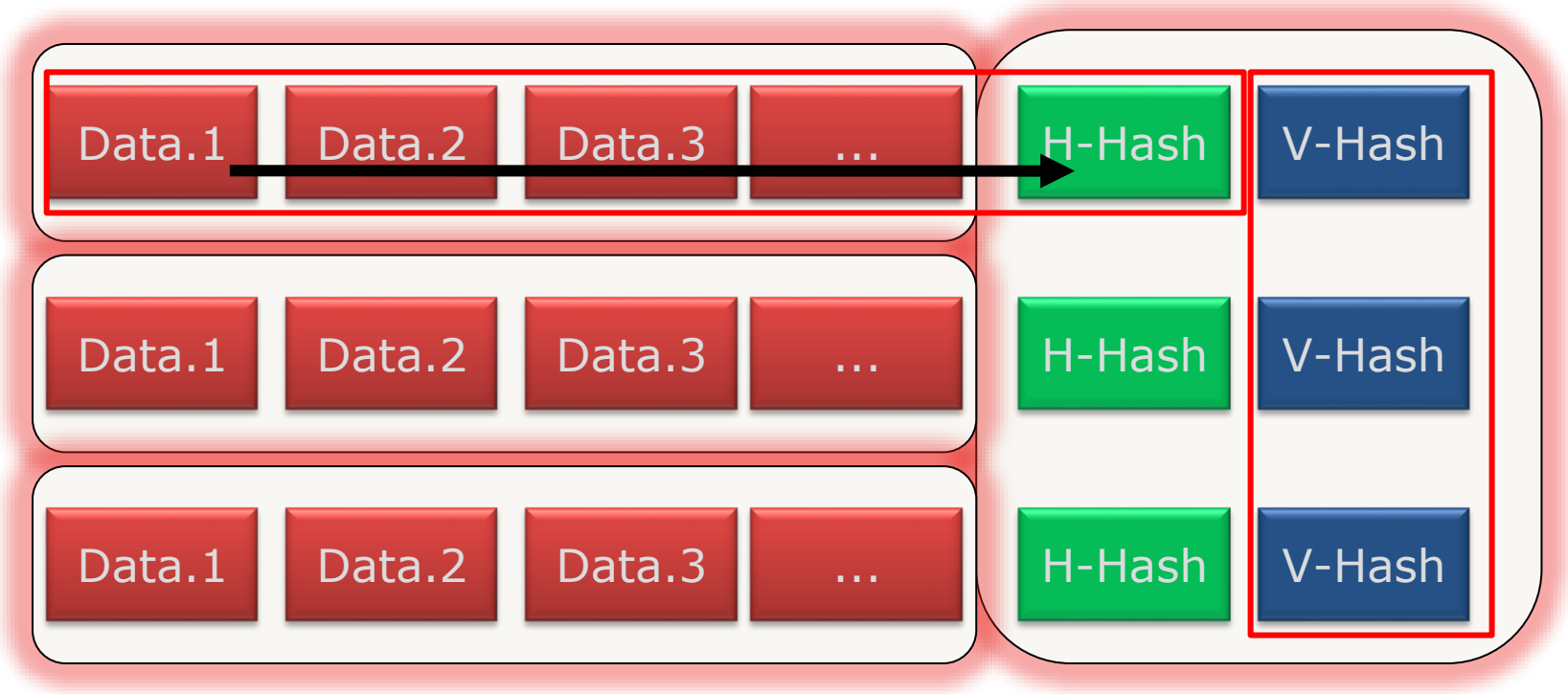
- DML operations are much more frequent in the DB;
  - INSERT
  - UPDATE
  - DELETE
- To gather information on the event types are used AFTER triggers;
  - Unlike the DDL, DML triggers were embroidered in the table;
- Performance issues;
  - No need to put trigger on all tables;
  - The problem can be solved by generic CLR triggers

Dilemma

- Seen **so far** as these methods secured enough to be used?

Anti-Tampering

- One of the custom methods



Tamper detection model

- One usage scenario

